EMPYREAN

An additive ambient synthesizer: twenty thousand partials per voice, on the graphics card

Version 0.9.0 · 151 parameters · 1024 presets · manual built 12.10.2026



Contents

1. The instrument
2. The cloud
3. Motion and landscapes
4. Filter and envelope
5. Tuning
6. The conductor
7. Space
8. Echoes, rooms and the modal hall
9. Modulation
10. Presets
11. Family jam and headset
12. Playing
13. MIDI
14. The presets
15. What it needs
16. License

The instrument

An additive ambient synthesizer: every voice is twenty thousand sine waves, computed on the graphics card. Each harmonic of a note is not one sine but a cloud of slightly detuned ones, every partial with its own slow drift, its own shimmer and its own place in the stereo field.

It plays by itself or under your hands. The conductor (Autoplay) is Noctuary's: a chord in just intonation that changes one voice at a time, minutes apart, in the spirit of Robert Rich's sleep concerts. Press Play in the header (or Space in the standalone) to start or stop it; play the keyboard over it, or switch it off and play alone.

The picture under the header shows every sounding voice as its spectrum of clouds: green for what you play, violet for the conductor, pale blue for the second conductor. The brighter a cloud, the louder and nearer it is. On the right, the landscape the Aurora motion reads, with a line where the voices are in its loop.

The cloud



Spectrum shapes the harmonics: Tilt makes the upper ones fall away, Brightness closes a window over them, Odd / Even thins one kind, Inharmonic stretches them as a stiff string does. The voice keeps its level whatever the shape: these colour the sound, they do not make it louder.

Cloud sets how wide each harmonic's cloud is (in cents, and how it widens towards the top), how far every partial drifts in pitch and how far its level shimmers, and how far the partials scatter across the stereo field. Purity (Tuning) narrows the clouds too, so that just intervals can lock.

Air adds a band of noise made of partials around a multiple of the fundamental: Centre, Width and Flicker (how fast its partials quiver -- which is what makes them heard as noise).

CLOUD

Control	Key	Range	Default
Partials	`partials`	200 .. 20000	20000
Spectral Tilt	`tilt`	0.3 .. 3	1.2
Brightness	`brightness`	0 .. 1	1
Odd / Even	`odd_even`	-1 .. 1	0
Inharmonic	`inharmonic`	0 .. 1	0
Width	`cloud_width`	0 .. 100 ct	25 ct
Width Slope	`width_slope`	0 .. 1	0.3
Drift	`drift`	0 .. 30 ct	3 ct

Drift Rate	`drift_rate`	0.01 .. 2 Hz	0.1 Hz
Shimmer	`shimmer`	0 .. 1	0.4
Shimmer Rate	`shimmer_rate`	0.01 .. 2 Hz	0.15 Hz
Stereo	`stereo`	0 .. 1	1

AIR

Control	Key	Range	Default
Air	`air`	0 .. 1	0
Centre	`air_center`	1 .. 12 x	3 x
Width	`air_width`	50 .. 2400 ct	600 ct
Flicker	`air_jitter`	0.5 .. 40 Hz	8 Hz

Motion and landscapes



Every partial has envelopes of its own, on three lanes: loudness, pitch and place. A motion sets them all at once: Random (every partial for itself, a fine glitter), Wave (a wave climbing through the spectrum), Pendulum (each harmonic pulses a little faster than the one below, over a bass that stays), Breathe (loudness, place and pitch travel through each cloud), Glissando (every partial glides into its note from up to fifteen semitones away), Aurora (the partials read a landscape).

A landscape is time against frequency: every partial reads the column at its own frequency. Aurora, Shepard (bands climbing for ever), Tides, Rain (short glints) and Choir (the formants of u-o-a-e-i, the vowel turning slowly). Rate and Depth scale whatever the motion does; Depth 0 leaves the bare cloud.

The field moves the partials as a whole, on top of any motion. Scale pulls every partial towards the nearest pitch of the scale: a wide or stretched cloud gathers into the key and keeps only a narrow shimmer. Swarm lets a few attractors (Attractors) wander through the spectrum on chaotic paths (Wander is how fast) and gathers the partials within their reach (Radius); Scatter pushes the partials away from them instead, gaps moving through the sound like dust parted by a wind. Pull is how fast the partials may move, Spring how strongly they are drawn back to their own places. The partials keep their own phases while they gather, so the level stays.

Dispersion makes each partial's clock late by its pitch. Positive, the low ones run up to that many seconds late: an attack or a glint falls from the top as a cascade, like a drop on ice. Negative, the high ones are late and it rises. On a held sound it is heard where the sound changes: attacks, releases, the glints of the landscapes.

MOTION

Control	Key	Range	Default
Motion	`motion`	None, Random, Wave, Pendulum, Breathe, Glissando, Aurora	Breathe
Rate	`motion_rate`	0.1 .. 4	1
Depth	`motion_depth`	0 .. 1	1
Landscape	`landscape`	Aurora, Shepard, Tides, Rain, Choir	Aurora
Dispersion	`dispersion`	-1.5 .. 1.5 s	0 s

FIELD

Control	Key	Range	Default
Field	`field_mode`	Off, Scale, Swarm, Scatter	Off
Pull	`field_strength`	0 .. 1	0.5
Radius	`field_radius`	5 .. 600 ct	60 ct
Spring	`field_spring`	0 .. 1	0.2
Wander	`field_rate`	0.005 .. 1	0.05
Attractors	`field_count`	1 .. 8	4

Filter and envelope



The filter is read as a frequency response at every partial: LP 6, LP 12, LP 24, HP 12, BP 12, Notch, Peak and Formant (whose cutoff picks the vowel). Env Amount moves the cutoff with the voice's envelope, in octaves; Key Track follows the note; Drift lets the cutoff wander on its own. MIDI controller 74 moves the cutoff.

The envelope is every voice's: Attack, Decay, Sustain, Release, in seconds to minutes.

FILTER

Control	Key	Range	Default
Model	`filter_model`	Off, LP 6, LP 12, LP 24, HP 12, BP 12, Notch, Peak, Formant	Off
Cutoff	`cutoff`	40 .. 18000 Hz	2500 Hz
Resonance	`resonance`	0 .. 1	0.15
Env Amount	`filter_env`	-4 .. 4 oct	0 oct
Key Track	`keytrack`	0 .. 1	0.5
Drift	`filter_drift`	0 .. 1	0.3

ENVELOPE

Control	Key	Range	Default
Attack	`attack`	0.01 .. 60 s	6 s
Decay	`decay`	0.01 .. 60 s	4 s
Sustain	`sustain`	0 .. 1	1

Release	`release`	0.05 .. 120 s	12 s

Tuning



Scale chooses the tuning: equal temperament, the just scales, the harmonic series, slendro, Bohlen-Pierce, or Timbre -- the scale the cloud's own spectrum asks for (Sethares). Root is its tonic; Keys maps the keyboard (snapped to twelve keys, or one key per degree). Purity blends equal temperament (0) with the scale (1). Tide leans the whole instrument by a few cents on a curve of minutes.

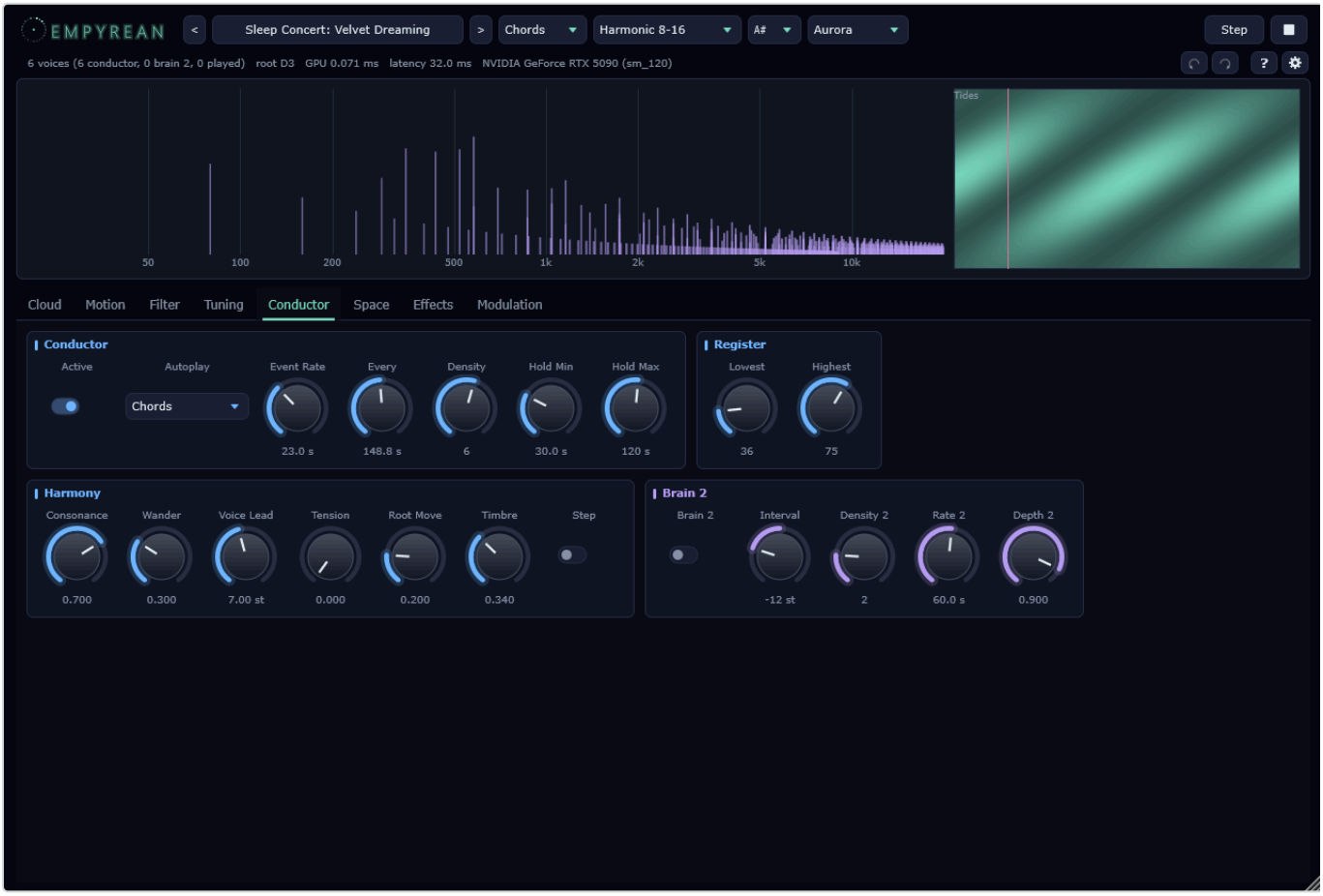
Self-Tune lets a chord tune itself. Where a harmonic of one voice nearly meets a harmonic of another (within Reach), the two voices pull together until they no longer beat: an equal-tempered major third settles to a pure one within seconds, and gives way again when the chord changes. Anchor draws every voice back towards its pitch in the scale.

TUNING

Control	Key	Range	Default
Scale	`scale`	12-TET, JI Major (Ptolemy), JI Minor, JI 7-limit, Pythagorean, JI Pentatonic, Harmonic 8-16, Subharmonic 16-8, Slendro (JI), Bohlen-Pierce (JI), Otonality 1-11, User (Scala), Timbre (Sethares)	JI 7-limit
Root	`root`	C, C#, D, D#, E, F, F#, G, G#, A, A#, B	D
Keys	`keymap`	Snap to 12 keys, Consecutive degrees	Snap to 12 keys
A4	`ref_pitch`	415 .. 466 Hz	440 Hz
Purity	`purity`	0 .. 1	1
Tide	`tide`	0 .. 30 ct	0 ct

Tide Period	`tide_period`	1 .. 60 min	12 min
Self-Tune	`tune_gravity`	0 .. 1	0
Reach	`tune_reach`	5 .. 60 ct	30 ct
Anchor	`tune_anchor`	0 .. 1	0.2

The conductor



Free lets notes come and go on their own timers. Chords (Autoplay) keeps the chord full and exchanges one voice at a time, every so many seconds (Every) or now (Step): the new note is the one that sits best against the voices that stay, within Voice Lead semitones. Tension allows notes that lean; Root Move lets the harmony travel. Timbre judges consonance from the cloud's own spectrum.

Brain 2 is a second conductor in the background: its root follows the first one's plus Interval, it plays slowly (Rate 2), few notes (Density 2), deep in the space (Depth 2).

CONDUCTOR

Control	Key	Range	Default
Active	`brain_on`	off / on	on
Autoplay	`auto_mode`	Free, Chords	Chords
Event Rate	`brain_rate`	2 .. 300 s	25 s
Every	`auto_rate`	2 .. 900 s	45 s
Density	`brain_density`	1 .. 10	5
Hold Min	`brain_hold_min`	5 .. 600 s	30 s
Hold Max	`brain_hold_max`	5 .. 600 s	120 s
Lowest	`brain_low`	24 .. 96	36
Highest	`brain_high`	24 .. 108	79
Consonance	`brain_consonance`	0 .. 1	0.7

Wander	`brain_wander`	0 .. 1	0.3
Voice Lead	`auto_lead`	1 .. 24 st	7 st
Tension	`auto_tension`	0 .. 1	0
Root Move	`auto_root_move`	0 .. 1	0.2
Timbre	`brain_timbre`	0 .. 1	0
Step	`auto_step`	off / on	off
Brain 2	`brain2_on`	off / on	off
Interval	`brain2_interval`	-24 .. 24 st	-12 st
Density 2	`brain2_density`	1 .. 6	2
Rate 2	`brain2_rate`	2 .. 300 s	60 s
Depth 2	`brain2_depth`	0 .. 1	0.9

Space



Every note stands on a plane between the ear (0) and the far background (1): the conductor puts some notes near and most deep (Depth), played notes stand at Keys Depth. A far note is quieter, darker and heard only through the far reverb. Breath lets every voice's plane wander; Doppler raises a voice that comes closer. Presence lifts the 2-5 kHz band on the near plane; Pad Low Cut leaves the bottom free.

The far reverb (Level, Size, Decay, Damping) is Noctuary's, with its Diffusion (the swell before the hall), Pre-Delay, Unmask (the hall steps aside, band by band, wherever the near plane sounds), its four networks (Far Mode: Colourless rings least, Rotating lets the tail drift) and Freeze.

Binaural (for headphones) places every partial of the near plane at a point of its own round the head: the head's shadow, the folds of the ear and the time the sound takes round the head, worked out for each partial at its own frequency. Surround spreads the places from the front half to all round, Height lifts and lowers them, Orbit turns the whole cloud round the listener. The far plane stays as it was, heard through the far reverb; the master's width and mono bass step aside while Binaural is on. Master: the level, the bass made mono below Bass Mono, and the width.

SPACE

Control	Key	Range	Default
Depth	`depth`	0 .. 1	0.6
Keys Depth	`keys_depth`	0 .. 1	0.2
Far Level	`far_level`	0 .. 1	0.8

Far Size	far_size	0.5 .. 3	2
Far Decay	far_decay	1 .. 90 s	25 s
Far Damping	far_damp	0 .. 1	0.5
Presence	presence	0 .. 6 dB	0 dB
Pad Low Cut	pad_low_cut	0 .. 300 Hz	0 Hz
Breath	breath	0 .. 1	0
Breath Rate	breath_rate	0.005 .. 0.2 Hz	0.03 Hz
Doppler	doppler	0 .. 1	0
Diffusion	far_diffuse	0 .. 1	0.5
Pre-Delay	far_predelay	0 .. 500 ms	20 ms
Unmask	far_unmask	0 .. 1	0.3
Far Mode	far_mode	Classic, Scattering, Colourless, Rotating	Colourless
Far Freeze	far_freeze	off / on	off

MASTER

Control	Key	Range	Default
Master	master_gain	-40 .. 12 dB	0 dB
Bass Mono	bass_mono	40 .. 300 Hz	150 Hz
Width	stereo_width	0 .. 2	1.2

BINAURAL

Control	Key	Range	Default
Binaural	binaural	off / on	off
Surround	binaural_span	0 .. 1	0.5
Height	binaural_height	0 .. 1	0.3
Orbit	binaural_orbit	-0.5 .. 0.5 Hz	0 Hz

Echoes, rooms and the modal hall



Echo is Noctuary's: two delays whose times glide and wander, a little different in each ear. Cross takes each side's feedback partly from the other, Damping darkens the loop, Absorb narrows it to a band as the feedback rises (so long echoes drown in fog instead of merely getting quieter), and Duck darkens it while the voices are loud, so a new note does not have to fight the last one's tail. To Far sends the echoes into the far plane, where they recede into the far reverb. Echo 2 runs after the first and hears its echoes: longer chains.

Room gives the near plane a place to stand: the early reflections of a room (Room Size, Absorb), arriving from where the voices stand, and a small reverb (Near Reverb, Near Decay, Near Damping).

The modal hall is a reverb made the way the instrument is made: sixty-five thousand resonances on the graphics card, each a sine that dies away, woken by everything the voices play. Decay is the time it takes to fall silent (60 dB) in the low range; Damping lets the upper octaves die sooner. Tuned puts that share of the resonances on the pitches of the scale, and Spread scatters them a few cents around their notes: with Tuned up, the hall rings in key, like the sympathetic strings of a viola d'amore, and retunes without a click when the scale or its root changes. Drift lets every resonance wander by a few cents, Width gives each its own place in each ear, Lowest keeps the bottom free. Breath lets every resonance's decay wander slowly, so the hall's colour moves while it rings. Paint lets the hall be heard through the landscape: its resonances light up where the landscape is bright, so the tail draws the aurora, the rain, the tides. Freeze holds the hall as it is, its input shut: a chord left hanging in the air.

The spectral echo takes the sound apart into five hundred bands and gives each a delay line of its own. Tilt brings the high bands back later than the low ones (or earlier), Scatter gives every band a time of its own, Feedback and Damping set how long and how dark the repeats are, To Far how much of them recedes. An attack comes back as a cascade, a chord as a spreading shimmer.

Memory keeps a ghost of every note let go. It rises over Bloom once the note has gone, fades over Memory Time, and while it fades it widens, drifts and darkens (Blur) and recedes into the far plane (Sink): the harmony of the last minutes stands faintly behind the present one.

ECHO

Control	Key	Range	Default
Echo	`delay_mix`	0 .. 1	0
Time L	`delay_time_l`	0.02 .. 4 s	0.75 s
Time R	`delay_time_r`	0.02 .. 4 s	1.13 s
Feedback	`delay_feedback`	0 .. 0.95	0.45
Cross	`delay_cross`	0 .. 1	0.35
Damping	`delay_damp`	0 .. 1	0.45
Absorb	`delay_absorb`	0 .. 1	0.3
Duck	`delay_duck`	0 .. 1	0.3
To Far	`delay_to_far`	0 .. 1	0.4
Echo 2	`delay2_mix`	0 .. 1	0
Time L 2	`delay2_time_l`	0.05 .. 4 s	2.4 s
Time R 2	`delay2_time_r`	0.05 .. 4 s	3.1 s
Feedback 2	`delay2_feedback`	0 .. 0.95	0.55
Damping 2	`delay2_damp`	0 .. 1	0.6
To Far 2	`delay2_to_far`	0 .. 1	0.6

ROOM

Control	Key	Range	Default
Early	`early_level`	0 .. 1	0.2
Room Size	`early_size`	2 .. 40 m	14 m
Absorb	`early_absorb`	0 .. 1	0.4
Near Reverb	`near_mix`	0 .. 1	0.1
Near Decay	`near_decay`	0.2 .. 8 s	1.8 s
Near Damping	`near_damp`	0 .. 1	0.5

MODAL

Control	Key	Range	Default
Modal Hall	`modal_level`	0 .. 1	0
Decay	`modal_decay`	0.5 .. 60 s	8 s
Damping	`modal_damp`	0 .. 1	0.35

Tuned	`modal_tune`	0 .. 1	0.5
Spread	`modal_spread`	0 .. 50 ct	6 ct
Drift	`modal_drift`	0 .. 30 ct	4 ct
Width	`modal_width`	0 .. 1	1
Lowest	`modal_low`	30 .. 400 Hz	80 Hz
Freeze	`modal_freeze`	off / on	off
Breath	`modal_breath`	0 .. 1	0
Paint	`modal_paint`	0 .. 1	0

SPECTRAL ECHO

Control	Key	Range	Default
Spectral Echo	`spec_mix`	0 .. 1	0
Time	`spec_time`	0.05 .. 4 s	0.8 s
Tilt	`spec_tilt`	-1 .. 1	0.5
Scatter	`spec_scatter`	0 .. 1	0.3
Feedback	`spec_feedback`	0 .. 0.95	0.5
Damping	`spec_damp`	0 .. 1	0.4
To Far	`spec_to_far`	0 .. 1	0.4

MEMORY

Control	Key	Range	Default
Memory	`memory_level`	0 .. 1	0
Memory Time	`memory_time`	10 .. 300 s	60 s
Bloom	`memory_bloom`	0.2 .. 20 s	4 s
Blur	`memory_blur`	0 .. 1	0.5
Sink	`memory_sink`	0 .. 1	0.6

Modulation



Four LFOs, two envelopes and four macros move the instrument's knobs through a matrix of sixteen rows: a source, a target, a depth. The LFOs are slow by default (their rates step by the golden ratio, so they never line up) and have no hard edges: Square and Ramp glide through their jumps in a few percent of a cycle, Random is a drifting curve, Steps glides from one value to the next. The envelopes are shapes from the motions' bank, played from the start whenever a note arrives into silence; Rate sets how fast they run.

The macros are knobs without a place of their own: they mean what the matrix says. Every factory preset gives them the same meaning, so that a controller mapped to them plays every preset: Macro 1 the motion (Depth and Rate), Macro 2 the brightness or the filter, Macro 3 the air, Macro 4 the depth in the space. The mod wheel, channel pressure, the velocity of the last note and a slow random curve are sources too.

Depth is measured in the knob's own travel: 0.5 moves a knob by half its range up and down, whatever its unit. A knob that is modulated shows a ring around it where its value is now.

MODULATION

Control	Key	Range	Default
LFO 1	`lfo1_shape`	Sine, Triangle, Ramp Up, Ramp Down, Square, Random, Steps	Sine
LFO 1 Rate	`lfo1_rate`	0.001 .. 20 Hz	0.05 Hz
LFO 2	`lfo2_shape`	Sine, Triangle, Ramp Up, Ramp Down, Square, Random, Steps	Triangle
LFO 2 Rate	`lfo2_rate`	0.001 .. 20 Hz	0.0809 Hz

LFO 3	`lfo3_shape`	Sine, Triangle, Ramp Up, Ramp Down, Square, Random, Steps	Random
LFO 3 Rate	`lfo3_rate`	0.001 .. 20 Hz	0.1309 Hz
LFO 4	`lfo4_shape`	Sine, Triangle, Ramp Up, Ramp Down, Square, Random, Steps	Steps
LFO 4 Rate	`lfo4_rate`	0.001 .. 20 Hz	0.2118 Hz
Env 1	`env1_shape`	Swell, Pulse, Bloom, Flicker, Wobble, Rise In, Fall In, Saw Up	Bloom
Env 1 Rate	`env1_rate`	0.02 .. 8 x	0.25 x
Env 2	`env2_shape`	Swell, Pulse, Bloom, Flicker, Wobble, Rise In, Fall In, Saw Up	Swell
Env 2 Rate	`env2_rate`	0.02 .. 8 x	0.1 x
Macro 1	`macro1`	0 .. 1	0
Macro 2	`macro2`	0 .. 1	0
Macro 3	`macro3`	0 .. 1	0
Macro 4	`macro4`	0 .. 1	0

Presets

The bank holds 1024 presets in sixteen groups of 64: Sleep Concert, Choirs, Glass, Bells, Strings, Organs, Air & Noise, Deep Space, Glissandi, Pendulums, Waves, Shepard & Tides, Shimmer, Dark Drones, Microtonal and Keys. The arrows in the header step through them; a click on the name opens the list by group. Every preset was measured by its loudest second and brought to the same level, so stepping through them does not jump.

MIDI program change chooses a preset within the group that bank select (controller 0) names. A preset sets every knob and the matrix; your MIDI-learn bindings stay.

Family jam and headset

The instruments of the family (Noctuary, Ephemeris, Phosphene, Totality, Parhelion and this one) can play together on one computer or a local network: Settings > Family jam. A follower keeps its conductor's root on the leader's (the nearest note of that pitch class, in its own tuning) and opens or closes with the energy of the leader's section; a leader tells the others its root and scale.

The Meta Quest app has no graphics card to compute twenty thousand partials per voice, so the instrument stays on this computer and the headset sends only its hands (bridge mode): Settings > Headset. The left hand's height is the brightness (Macro 2), the right hand's the motion (Macro 1); the pinches play and stop the conductor, step it, bring Brain 2 in and out, and both hands pinched choose the next preset.

Playing

The keyboard plays over the conductor. The hold pedal (controller 64) keeps notes sounding until it comes up. Right-click any control for MIDI learn (move a controller to bind it), to forget a binding or to return to the default.

Everything you hear is delayed by the same few milliseconds (the engine renders ahead of the audio so the graphics card can never make it stutter); a host compensates for it.

MIDI

Message	What it does
Notes	play over the conductor (velocity: the note's level, and a source of the matrix)
Controller 0, then program change	a preset: the bank names the group (0 .. 15), the program the preset in it (0 .. 63)
Controller 1 (mod wheel)	a source of the matrix; the factory presets open the filter or the brightness with it
Controller 64 (hold pedal)	keeps notes sounding until it comes up
Controller 74	the filter's cutoff
Channel or key pressure	a source of the matrix; the factory presets lift the presence with it
All notes off, all sound off	every voice is released
Any other controller	MIDI learn: right-click a control, move the controller

The presets

0 · SLEEP CONCERT

0 Lunar Hall
1 Drowsy Hours
2 Velvet Harbour
3 Drowsy Cradle
4 Velvet Tide
5 Somnolent Hall
6 Drowsy Lantern
7 Slow Hours
8 Lunar Vigil
9 Slow Harbour
10 Drowsy Tide
11 Lunar Harbour
12 Deep Harbour
13 Nocturnal Tide
14 Drowsy Vigil
15 Deep Hall
16 Velvet Hall
17 Somnolent Vigil
18 Lunar Dreaming
19 Lunar Hours
20 Lunar Tide
21 Velvet Hours
22 Somnolent Cradle
23 Hushed Hours
24 Hushed Dreaming
25 Somnolent Tide
26 Drowsy Harbour
27 Deep Hours
28 Hushed Vigil
29 Deep Cradle
30 Deep Lantern
31 Lunar Cradle
32 Somnolent Harbour
33 Slow Tide
34 Hushed Lantern
35 Slow Vigil
36 Nocturnal Harbour
37 Slow Dreaming
38 Somnolent Lantern
39 Nocturnal Cradle
40 Deep Vigil
41 Velvet Cradle
42 Deep Tide
43 Velvet Lantern
44 Slow Hall
45 Nocturnal Hours
46 Hushed Hall
47 Slow Cradle
48 Drowsy Hall
49 Nocturnal Hall
50 Deep Dreaming
51 Nocturnal Dreaming
52 Nocturnal Vigil
53 Hushed Cradle
54 Lunar Lantern
55 Nocturnal Lantern
56 Hushed Tide
57 Somnolent Hours
58 Drowsy Dreaming
59 Hushed Harbour
60 Somnolent Dreaming
61 Velvet Vigil
62 Slow Lantern
63 Velvet Dreaming

1 · CHOIRS

0 Turning Chant
1 Vowel Choir
2 Cathedral Chant
3 Vowel Litany
4 Hollow Chant
5 Breathing Hymn
6 Cathedral Litany
7 Silent Aria
8 Breathing Voices
9 Turning Choir
10 Pale Hymn
11 Vowel Psalm
12 Vowel Chant
13 Hollow Aria
14 Distant Chorus
15 Pale Litany
16 Pale Psalm
17 Distant Psalm
18 Pale Aria
19 Silent Chorus
20 Silent Hymn
21 Hollow Choir
22 Breathing Litany
23 Cathedral Psalm
24 Pale Chorus
25 Silent Litany
26 Cathedral Choir
27 Vowel Voices
28 Distant Hymn
29 Vowel Hymn
30 Hollow Litany
31 Vowel Chorus
32 Silent Voices
33 Hollow Chorus
34 Turning Litany
35 Distant Chant
36 Pale Choir
37 Hollow Voices
38 Breathing Chorus
39 Turning Psalm
40 Breathing Aria
41 Turning Voices
42 Cathedral Aria
43 Distant Voices
44 Distant Choir
45 Breathing Chant
46 Silent Psalm
47 Silent Choir
48 Cathedral Voices
49 Pale Voices
50 Breathing Psalm
51 Vowel Aria
52 Pale Chant
53 Cathedral Hymn
54 Hollow Hymn
55 Turning Hymn
56 Cathedral Chorus
57 Breathing Choir
58 Distant Litany
59 Turning Aria
60 Hollow Psalm
61 Silent Chant
62 Turning Chorus
63 Distant Aria

2 · GLASS

0 Thin Spire
1 Frosted Spire
2 Cut Spire
3 Crystal Facet
4 Lucid Harmonica
5 Cold Facet

6 Frosted Vial
7 Clear Pane
8 Brittle Vial
9 Frosted Glass
10 Thin Glass
11 Brittle Prism
12 Thin Lens
13 Cold Spire
14 Brittle Pane
15 Brittle Spire
16 Brittle Facet
17 Brittle Harmonica
18 Frosted Harmonica
19 Lucid Spire
20 Crystal Vial
21 Clear Harmonica
22 Cut Glass
23 Thin Harmonica
24 Cold Prism
25 Lucid Vial
26 Clear Lens
27 Cold Glass
28 Cold Pane
29 Cut Prism
30 Thin Vial
31 Thin Facet
32 Cut Harmonica
33 Crystal Prism
34 Cold Harmonica
35 Lucid Facet
36 Cut Lens
37 Thin Pane
38 Clear Facet
39 Cold Vial
40 Lucid Glass
41 Crystal Pane
42 Crystal Harmonica
43 Frosted Prism
44 Cut Facet
45 Frosted Facet
46 Lucid Lens
47 Clear Spire
48 Frosted Pane
49 Crystal Glass
50 Cut Pane
51 Brittle Lens
52 Frosted Lens
53 Crystal Spire
54 Crystal Lens
55 Brittle Glass
56 Lucid Pane
57 Cut Vial
58 Thin Prism
59 Clear Glass
60 Clear Prism
61 Clear Vial
62 Lucid Prism
63 Cold Lens

3 · BELLS

0 Iron Tower
1 Iron Bowl
2 Far Carillon
3 Temple Toll
4 Temple Carillon
5 Sunken Gong
6 Tolling Tower
7 Iron Toll
8 Temple Gong
9 Iron Bell
10 Tibetan Gong
11 Sunken Chime
12 Struck Carillon
13 Tolling Peal
14 Tibetan Chime
15 Far Bell
16 Bronze Peal
17 Temple Tower
18 Iron Carillon
19 Struck Bowl
20 Tibetan Bowl
21 Struck Toll
22 Iron Gong
23 Tibetan Toll
24 Temple Chime
25 Struck Bell
26 Far Peal
27 Bronze Toll
28 Far Gong
29 Tibetan Tower
30 Temple Bowl
31 Tibetan Peal
32 Temple Bell
33 Bronze Bell
34 Sunken Toll
35 Far Chime
36 Bronze Carillon
37 Sunken Peal
38 Tolling Bell
39 Tolling Chime
40 Bronze Bowl
41 Tolling Carillon
42 Sunken Tower
43 Tibetan Carillon
44 Far Bowl
45 Bronze Tower
46 Tibetan Bell
47 Tolling Gong
48 Sunken Bell
49 Struck Peal
50 Bronze Chime
51 Far Tower
52 Iron Chime
53 Bronze Gong
54 Struck Gong
55 Tolling Bowl
56 Sunken Bowl
57 Struck Chime
58 Struck Tower
59 Iron Peal
60 Temple Peal
61 Far Toll
62 Sunken Carillon
63 Tolling Toll

4 · STRINGS

0 Sustained Strings
1 Sustained Fibres
2 Sustained Viols
3 Amber Viols
4 Sustained Cellos
5 Amber Ensemble
6 Muted Section
7 Warm Strings
8 Amber Bows
9 Warm Ensemble
10 Silken Cellos
11 Bowed Cellos
12 Long Cellos
13 Silken Ensemble
14 Warm Cellos
15 Silken Bows
16 Warm Bows
17 Long Bows
18 Muted Viols
19 Bowed Fibres
20 Sustained Quartet
21 Warm Viols
22 Sustained Bows
23 Muted Quartet
24 Gliding Ensemble
25 Long Fibres
26 Sustained Section

27 Long Ensemble
28 Long Strings
29 Gliding Fibres
30 Muted Ensemble
31 Sustained Ensemble
32 Warm Section
33 Long Section
34 Bowed Quartet
35 Muted Bows
36 Silken Strings
37 Bowed Viols
38 Amber Quartet
39 Bowed Strings
40 Silken Fibres
41 Amber Strings
42 Bowed Section
43 Warm Fibres
44 Amber Section
45 Silken Viols
46 Gliding Quartet
47 Amber Cellos
48 Bowed Ensemble
49 Gliding Viols
50 Warm Quartet
51 Silken Quartet
52 Amber Fibres
53 Long Viols
54 Silken Section
55 Gliding Strings
56 Muted Strings
57 Gliding Cellos
58 Gliding Section
59 Muted Cellos
60 Muted Fibres
61 Gliding Bows
62 Bowed Bows
63 Long Quartet

5 · ORGANS

0 Mixture Flute
1 Pedal Principal
2 Ancient Principal
3 Mixture Diapason
4 Open Principal
5 Reed Diapason
6 Pedal Nave
7 Stopped Organ
8 Open Celeste
9 Ancient Nave
10 Stopped Bourdon
11 Ancient Pipes
12 Chapel Organ
13 Chapel Pipes
14 Pedal Organ
15 Open Flute
16 Ancient Bourdon
17 Mixture Bourdon
18 Reed Pipes
19 Reed Organ
20 Stopped Pipes
21 Flue Pipes
22 Mixture Celeste
23 Reed Bourdon
24 Chapel Nave
25 Open Organ
26 Pedal Bourdon
27 Reed Nave
28 Open Bourdon
29 Chapel Celeste
30 Ancient Celeste
31 Flue Diapason
32 Stopped Principal
33 Reed Celeste
34 Flue Organ
35 Stopped Celeste
36 Pedal Pipes
37 Flue Principal
38 Stopped Flute
39 Pedal Celeste
40 Ancient Flute
41 Chapel Principal
42 Pedal Flute
43 Stopped Diapason
44 Reed Principal
45 Flue Nave
46 Flue Bourdon
47 Mixture Nave
48 Stopped Nave
49 Open Nave
50 Reed Flute
51 Ancient Diapason
52 Chapel Flute
53 Flue Celeste
54 Flue Flute
55 Ancient Organ
56 Chapel Bourdon
57 Mixture Organ
58 Open Diapason
59 Chapel Diapason
60 Open Pipes
61 Mixture Pipes
62 Pedal Diapason
63 Mixture Principal

6 · AIR & NOISE

0 Grain Spray
1 Salt Hiss
2 Rustling Spray
3 Ash Breath
4 Rain Drift
5 Wind Weather
6 Dusty Hiss
7 Rustling Weather
8 Dusty Weather
9 Dusty Spray
10 Grain Veil
11 Wind Hiss
12 Ash Field
13 Grain Weather
14 Rustling Hiss
15 Static Field
16 Rustling Veil
17 Salt Field
18 Salt Drift
19 Rain Mist
20 Salt Veil
21 Salt Weather
22 Rustling Drift
23 Static Drift
24 Static Veil
25 Static Weather
26 Static Mist
27 Ash Spray
28 Ash Weather
29 Grain Hiss
30 Ash Mist
31 Rustling Field
32 Rain Veil
33 Wind Spray
34 Dusty Drift
35 Dusty Field
36 Wind Drift
37 Grain Breath
38 Rain Weather
39 Ash Hiss
40 Dusty Breath
41 Rain Hiss
42 Rain Spray
43 Wind Breath
44 Rustling Breath
45 Wind Mist
46 Salt Spray
47 Static Breath

48 Grain Field
49 Dusty Veil
50 Rain Field
51 Salt Breath
52 Ash Drift
53 Rain Breath
54 Ash Veil
55 Grain Drift
56 Dusty Mist
57 Salt Mist
58 Static Spray
59 Wind Field
60 Rustling Mist
61 Static Hiss
62 Wind Veil
63 Grain Mist

7 · DEEP SPACE

0 Cosmic Drift
1 Dark Nebula
2 Orbital Corona
3 Stellar Abyss
4 Silent Nebula
5 Cosmic Abyss
6 Stellar Drift
7 Silent Abyss
8 Void Corona
9 Dark Corona
10 Cosmic Corona
11 Cosmic Nebula
12 Orbital Abyss
13 Event Expanse
14 Remote Expanse
15 Void Expanse
16 Void Horizon
17 Remote Station
18 Stellar Expanse
19 Silent Horizon
20 Orbital Horizon
21 Remote Nebula
22 Silent Corona
23 Stellar Horizon
24 Silent Station
25 Silent Drift
26 Dark Station
27 Orbital Nebula
28 Cosmic Expanse
29 Void Station
30 Dark Horizon
31 Event Signal
32 Event Station
33 Cosmic Horizon
34 Event Abyss
35 Orbital Drift
36 Void Drift
37 Dark Drift
38 Stellar Nebula
39 Silent Signal
40 Event Horizon
41 Orbital Station
42 Remote Drift
43 Cosmic Station
44 Remote Corona
45 Silent Expanse
46 Stellar Station
47 Remote Signal
48 Dark Expanse
49 Event Nebula
50 Orbital Signal
51 Remote Horizon
52 Dark Abyss
53 Void Signal
54 Void Abyss
55 Event Corona
56 Event Drift
57 Cosmic Signal
58 Void Nebula
59 Dark Signal
60 Remote Abyss
61 Stellar Corona
62 Stellar Signal
63 Orbital Expanse

8 · GLISSANDI

0 Arriving Birds
1 Gathering Choir
2 Swarming Chord
3 Falling Choir
4 Sliding Birds
5 Bending Birds
6 Bending Glissando
7 Rising Glissando
8 Rising Cloud
9 Sliding Choir
10 Gathering Swarm
11 Swarming Voices
12 Gathering Chord
13 Rising Lights
14 Gathering Cloud
15 Converging Birds
16 Sliding Lights
17 Converging Lights
18 Rising Swarm
19 Bending Lights
20 Swarming Birds
21 Bending Cloud
22 Converging Voices
23 Bending Choir
24 Converging Swarm
25 Rising Chord
26 Arriving Swarm
27 Converging Choir
28 Falling Lights
29 Swarming Choir
30 Converging Chord
31 Bending Chord
32 Arriving Glissando
33 Converging Glissando
34 Sliding Cloud
35 Swarming Swarm
36 Sliding Glissando
37 Rising Voices
38 Falling Swarm
39 Arriving Cloud
40 Arriving Voices
41 Converging Cloud
42 Rising Birds
43 Gathering Glissando
44 Gathering Lights
45 Bending Swarm
46 Falling Glissando
47 Swarming Cloud
48 Gathering Birds
49 Arriving Chord
50 Falling Cloud
51 Bending Voices
52 Sliding Chord
53 Rising Choir
54 Arriving Lights
55 Swarming Lights
56 Sliding Swarm
57 Falling Chord
58 Falling Birds
59 Arriving Choir
60 Sliding Voices
61 Falling Voices
62 Gathering Voices
63 Swarming Glissando

9 · PENDULUMS

0 Patient Pendulum
1 Patient Weights
2 Pacing Escapement
3 Clock Weights
4 Measured Pendulum
5 Swinging Clocks
6 Measured Gears
7 Pacing Metronome
8 Swinging Cycles
9 Measured Escapement
10 Slow Escapement
11 Measured Metronome
12 Pacing Weights
13 Steady Cycles
14 Swinging Escapement
15 Steady Weights
16 Clock Pendulum
17 Slow Clocks
18 Patient Escapement
19 Pacing Clocks
20 Steady Escapement
21 Turning Weights
22 Measured Orbits
23 Turning Gears
24 Pacing Pendulum
25 Slow Gears
26 Steady Clocks
27 Measured Clocks
28 Turning Escapement
29 Measured Weights
30 Patient Metronome
31 Patient Gears
32 Slow Pendulum
33 Turning Cycles
34 Steady Gears
35 Pacing Orbits
36 Patient Orbits
37 Turning Metronome
38 Swinging Gears
39 Clock Gears
40 Steady Pendulum
41 Steady Orbits
42 Steady Metronome
43 Turning Orbits
44 Slow Metronome
45 Clock Escapement
46 Swinging Weights
47 Swinging Orbits
48 Patient Clocks
49 Slow Orbits
50 Slow Cycles
51 Clock Orbits
52 Clock Metronome
53 Clock Clocks
54 Turning Pendulum
55 Slow Weights
56 Clock Cycles
57 Pacing Gears
58 Turning Clocks
59 Measured Cycles
60 Swinging Metronome
61 Pacing Cycles
62 Swinging Pendulum
63 Patient Cycles

10 · WAVES

0 Endless Shore
1 Swelling Current
2 Long Breakers
3 Endless Wake
4 Rising Shore
5 Rolling Shore
6 Endless Swell
7 Climbing Waves
8 Climbing Wake
9 Climbing Breakers
10 Climbing Shore
11 Climbing Current
12 Long Ocean
13 Calm Shore
14 Broken Ocean
15 Rising Swell
16 Rolling Swell
17 Rising Breakers
18 Rolling Surf
19 Long Current
20 Swelling Waves
21 Rolling Breakers
22 Rising Ocean
23 Rising Current
24 Calm Ocean
25 Broken Shore
26 Endless Surf
27 Calm Surf
28 Endless Ocean
29 Swelling Ocean
30 Long Waves
31 Rising Waves
32 Calm Breakers
33 Long Surf
34 Broken Breakers
35 Calm Waves
36 Rolling Waves
37 Rolling Current
38 Broken Swell
39 Long Shore
40 Broken Wake
41 Swelling Wake
42 Swelling Swell
43 Endless Waves
44 Calm Current
45 Broken Current
46 Long Swell
47 Long Wake
48 Rising Surf
49 Calm Swell
50 Broken Waves
51 Rising Wake
52 Rolling Wake
53 Climbing Ocean
54 Calm Wake
55 Endless Breakers
56 Climbing Swell
57 Climbing Surf
58 Endless Current
59 Swelling Breakers
60 Swelling Surf
61 Rolling Ocean
62 Broken Surf
63 Swelling Shore

11 · SHEPARD & TIDES

0 Climbing Ladder
1 Infinite Ladder
2 Climbing Tide
3 Ascending Flood
4 Ascending Moon
5 Circling Tide
6 Ascending Ladder
7 Endless Ebb
8 Tidal Ladder
9 Eternal Moon
10 Circling Ascent
11 Ascending Staircase
12 Infinite Flood
13 Circling Spiral
14 Infinite Staircase
15 Tidal Tide
16 Eternal Ascent
17 Endless Spiral

18 Eternal Ladder
19 Endless Ladder
20 Climbing Flood
21 Tidal Moon
22 Ascending Ebb
23 Climbing Ebb
24 Climbing Staircase
25 Endless Ascent
26 Eternal Flood
27 Circling Moon
28 Tidal Staircase
29 Spiral Flood
30 Ascending Ascent
31 Eternal Tide
32 Endless Staircase
33 Climbing Spiral
34 Circling Ebb
35 Infinite Moon
36 Spiral Ebb
37 Tidal Flood
38 Circling Ladder
39 Spiral Ascent
40 Endless Moon
41 Climbing Moon
42 Eternal Ebb
43 Ascending Tide
44 Circling Flood
45 Spiral Staircase
46 Eternal Spiral
47 Infinite Spiral
48 Spiral Moon
49 Endless Flood
50 Infinite Ascent
51 Infinite Ebb
52 Circling Staircase
53 Infinite Tide
54 Tidal Ascent
55 Ascending Spiral
56 Spiral Tide
57 Spiral Ladder
58 Tidal Spiral
59 Climbing Ascent
60 Spiral Spiral
61 Endless Tide
62 Eternal Staircase
63 Tidal Ebb

12 · SHIMMER

0 Trembling Dust
1 Sparkling Haze
2 Dappled Sheen
3 Glowing Aurora
4 Dappled Shimmer
5 Dappled Haze
6 Glinting Halo
7 Bright Dust
8 Sparkling Aurora
9 Silver Sheen
10 Flickering Shimmer
11 Glowing Halo
12 Dappled Halo
13 Glowing Haze
14 Bright Sheen
15 Trembling Light
16 Sparkling Dust
17 Flickering Sheen
18 Flickering Light
19 Glowing Sheen
20 Glinting Haze
21 Trembling Shimmer
22 Dappled Aurora
23 Glinting Aurora
24 Trembling Halo
25 Glinting Dust
26 Glinting Sheen
27 Glowing Shimmer
28 Sparkling Light
29 Dappled Veil
30 Sparkling Halo
31 Glowing Dust
32 Bright Shimmer
33 Flickering Haze
34 Flickering Dust
35 Flickering Veil
36 Sparkling Sheen
37 Glowing Light
38 Silver Shimmer
39 Bright Veil
40 Trembling Sheen
41 Silver Veil
42 Glinting Veil
43 Silver Aurora
44 Flickering Halo
45 Glinting Shimmer
46 Silver Haze
47 Silver Halo
48 Flickering Aurora
49 Glowing Veil
50 Bright Halo
51 Trembling Aurora
52 Trembling Haze
53 Silver Dust
54 Dappled Light
55 Sparkling Shimmer
56 Sparkling Veil
57 Glinting Light
58 Trembling Veil
59 Bright Aurora
60 Silver Light
61 Bright Light
62 Bright Haze
63 Dappled Dust

13 · DARK DRONES

0 Sombre Cavern
1 Heavy Drone
2 Black Cavern
3 Low Cavern
4 Slow Monolith
5 Heavy Monolith
6 Low Crypt
7 Heavy Engine
8 Black Basalt
9 Sombre Crypt
10 Buried Engine
11 Sombre Monolith
12 Brooding Drone
13 Buried Basalt
14 Slow Cavern
15 Low Furnace
16 Brooding Cavern
17 Sombre Basalt
18 Sombre Furnace
19 Slow Basalt
20 Subterranean Crypt
21 Heavy Undertow
22 Black Furnace
23 Black Crypt
24 Brooding Engine
25 Low Monolith
26 Black Drone
27 Buried Crypt
28 Subterranean Basalt
29 Buried Undertow
30 Black Engine
31 Buried Monolith
32 Sombre Engine
33 Heavy Cavern
34 Buried Cavern
35 Subterranean Furnace
36 Heavy Furnace
37 Subterranean Engine
38 Subterranean Cavern

39 Sombre Undertow
40 Black Monolith
41 Low Undertow
42 Buried Furnace
43 Subterranean Drone
44 Low Drone
45 Brooding Furnace
46 Heavy Crypt
47 Slow Engine
48 Low Basalt
49 Sombre Drone
50 Low Engine
51 Brooding Basalt
52 Black Undertow
53 Heavy Basalt
54 Brooding Undertow
55 Slow Crypt
56 Slow Furnace
57 Brooding Crypt
58 Subterranean Monolith
59 Slow Undertow
60 Subterranean Undertow
61 Buried Drone
62 Slow Drone
63 Brooding Monolith

14 · MICROTONAL

0 Eleventh Tuning
1 Eleventh Ratio
2 Tilted Tuning
3 Stretched Comma
4 Seventh Series
5 Tritave Gamelan
6 Eleventh Mode
7 Tilted Series
8 Crooked Gamelan
9 Tilted Lattice
10 Crooked Comma
11 Stretched Lattice
12 Foreign Comma
13 Tritave Scale
14 Tritave Mode
15 Bent Tuning
16 Stretched Ratio
17 Stretched Mode
18 Tilted Gamelan
19 Foreign Tuning
20 Eleventh Series
21 Bent Comma
22 Bent Gamelan
23 Tritave Tuning
24 Tritave Comma
25 Tritave Lattice
26 Tilted Comma
27 Foreign Series
28 Foreign Scale
29 Crooked Scale
30 Stretched Gamelan
31 Seventh Tuning
32 Tritave Series
33 Eleventh Scale
34 Tritave Ratio
35 Tilted Mode
36 Seventh Lattice
37 Tilted Ratio
38 Eleventh Gamelan
39 Seventh Gamelan
40 Bent Series
41 Foreign Gamelan
42 Seventh Comma
43 Foreign Mode
44 Bent Ratio
45 Stretched Series
46 Crooked Series
47 Seventh Mode
48 Crooked Tuning
49 Eleventh Comma
50 Bent Mode
51 Crooked Ratio
52 Seventh Scale
53 Crooked Mode
54 Foreign Lattice
55 Bent Scale
56 Seventh Ratio
57 Eleventh Lattice
58 Bent Lattice
59 Crooked Lattice
60 Stretched Tuning
61 Tilted Scale
62 Foreign Ratio
63 Stretched Scale

15 · KEYS

0 Close Notes
1 Felt Chords
2 Soft Notes
3 Felt Fingers
4 Played Fingers
5 Night Touch
6 Night Chords
7 Gentle Piano
8 Played Touch
9 Played Keys
10 Felt Hands
11 Played Chords
12 Close Chords
13 Touched Chords
14 Soft Pad
15 Soft Chords
16 Quiet Chords
17 Soft Touch
18 Night Notes
19 Felt Piano
20 Night Piano
21 Touched Keys
22 Night Fingers
23 Gentle Touch
24 Touched Touch
25 Felt Pad
26 Quiet Fingers
27 Felt Keys
28 Quiet Pad
29 Soft Hands
30 Played Piano
31 Close Keys
32 Gentle Hands
33 Soft Piano
34 Touched Fingers
35 Gentle Chords
36 Felt Touch
37 Played Notes
38 Gentle Keys
39 Touched Pad
40 Quiet Piano
41 Close Piano
42 Soft Fingers
43 Night Pad
44 Soft Keys
45 Gentle Fingers
46 Close Touch
47 Quiet Notes
48 Touched Piano
49 Gentle Notes
50 Close Hands
51 Played Pad
52 Close Pad
53 Touched Hands
54 Quiet Touch
55 Night Hands
56 Touched Notes
57 Gentle Pad
58 Close Fingers
59 Quiet Hands

60 Quiet Keys
61 Played Hands
62 Night Keys
63 Felt Notes

What it needs

Graphics card	NVIDIA GeForce RTX 20 or newer (Turing and later) with a current driver; the instrument computes every partial there and is silent without one, saying so in the status line
System	Windows 10 or 11, 64-bit, a processor with AVX2
Formats	standalone application and VST3 plug-in; the offline renderer empy_render for the command line
Latency	two audio blocks and ten milliseconds, reported to the host, which compensates for it

License

Empyrean is free software under the GNU Affero General Public License, version 3 (LICENSE.txt beside the program). It is built with JUCE and NVIDIA's CUDA; the CUDA runtime is linked into the program under NVIDIA's license for it.